# Conductor-gap panel: NIST K curves and standardized controls

2026-10-14

## Result in one page

This evidence-reduction round measures **43 certified prime conductor directions** across **20 finite-field trace/isogeny classes** (17 twist-deduplicated conductor geometries), comprising **38 distinct prime degrees**. Expanding every squarefree conductor gives **102 exact endomorphism-order levels** and **131 conditional adjacent-level placements**. Those 131 placements are repeated algebraic contexts, not 131 independent hardness observations. The machine-readable record is `panel.json`; the flattened direction, level, adjacency, construction, and measured-effect tables are the CSV files beside this report. The independent arithmetic and transcription receipt is `verification.json`, produced by `verify_panel.py`.

The main findings are:

1. **Large conductor gaps correlate with a derived direct-kernel point-field burden, not an ECDLP discount by themselves.** Across the 43 structural rows, `Spearman(log2 ell, log2(m*k)) = 0.9462` (derived, descriptive only), where `k` is the least exponent for which q-Frobenius acts as `+/-1` on an ell-kernel generator. Rows are clustered in only 20 classes, so this is not a significance test or a causal estimate; three twist pairs also share the same conductor geometry. The panel is purposive, all non-control gap rows use the `Q(sqrt(-7))` Koblitz mechanism, and no cross-panel discovery cost relationship was measured.
2. **The scalar-orbit degree `k` materially changes the ranking.** K-283's degree-1,697 direction needs kernel x-coordinates over an absolute degree 239,984 field, while K-163's larger degree-45,641 direction needs degree 3,260. Degree alone misses that 74-fold difference.
3. **The existing same-action level experiments do not show a multi-bit conductor effect.** All eight conditional edges at `m=37` and `m=41` have pointwise 95% intervals containing zero. The accepted `m=51`, degree-271 same-`tau^17` estimate is `+0.005763` bit with 95% interval `[-0.032170,+0.043374]`. A separate four-level `m=51` run has point estimates within 0.0264 bit, but its A/A gate failed, so its inferential labels are withheld.
4. **The three-to-four-bit scale appears only in the modeled action comparison.** A NIST K source has a derived ideal signed-Frobenius rho discount of 3.674 to 4.579 bits relative to negation. That belongs in the source baseline arm; it is not evidence that a conductor floor is weak. A claimed 3-bit weak endpoint must have a multiplicity-adjusted one-sided lower confidence bound above 3 bits after map, solve, recovery, failures, and verification are all charged.
5. **No NIST row yet reaches the transfer gate.** The five K curves provide 15 exact prime directions, but zero rows currently have a certified point-evaluable map, dual, subgroup transport, planted-scalar recovery, and fully charged transfer cost together. Therefore no NIST weakness assertion is eligible in this panel.

| requested scope | current status | evidence in this round |
|---|---|---|
| Dozens of vertical geometries | verified exact | 43 prime directions, 102 levels, 131 conditional placements |
| Per-level ECDLP effects | partial | admitted measurements only for `m=37`, `m=41`, and one `m=51` edge; a later `m=51` run is control-gated |
| K-163/K-283 map, dual, and transfer | blocked/unrun | no canonical selectable run and no complete transfer artifact |
| Horizontal K/P/B/`sect113r1` paths | not assessed | zero horizontal rows |

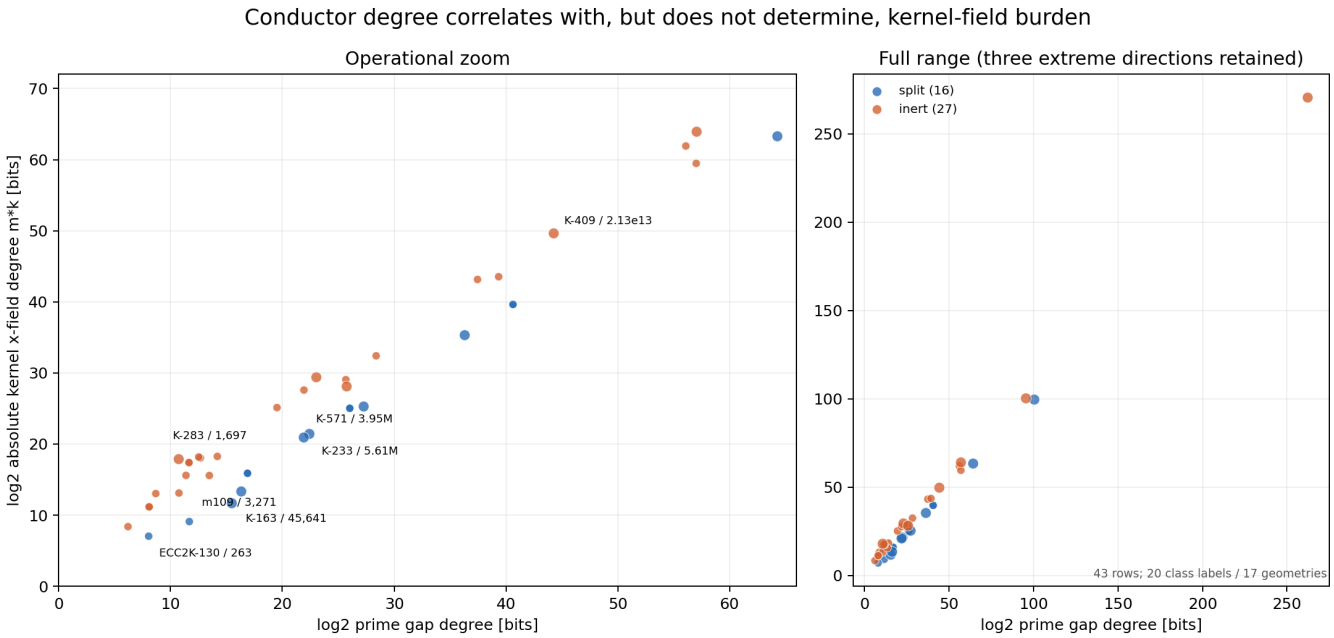



Figure 1: Prime conductor degree against the absolute kernel x-field degree. The left panel resolves the operational range; the right retains all extreme rows. Point count is 43, with 20 trace/isogeny-class labels and 17 twist-deduplicated geometries; no independence claim is made.

Editable vector: gap-landscape.svg; generator: build_panel.py.

## Objects and measurements

For each ordinary Koblitz crater

$$E_a : y^2 + xy = x^3 + ax^2 + 1 \quad \text{over } \mathbb{F}_{2^m},$$

the CM field is `K=Q(sqrt(-7))`, the named crater has `f_E=1`, and

$$t^2 - 4q = -7f_\pi^2.$$

Every prime `ell | f_pi` is a depth-one vertical conductor direction in this panel. If `tau^m=X+Y tau` and q-Frobenius is the scalar `c=X mod ell` on `E[ell]`, the direct kernel-point route uses

$$k = \min\{j > 0 : c^j = \pm 1 \pmod{\ell}\}, \qquad x(P) \in \mathbb{F}_{2^{mk}}.$$

The exact structural vector retained per direction is

$$(\ell, \log_2 \ell, \text{split type}, k, mk, (\ell-1)/2, \text{packed kernel bytes}, h(-7\ell^2)).$$

For squarefree `f_pi`, every divisor is an exact possible order level and

$$h(-7f_E^2) = \prod_{p \mid f_E} \left(p - \left(\frac{-7}{p}\right)\right).$$

The packed-byte column is the derived dense coefficient payload `ceil((((ell-1)/2)+1)*m/8)`. It is not an observed file size, memory peak, or evaluation cost. Likewise, existence of a conductor-lattice adjacency does not construct its rational map.

The panel keeps four notions separate:

- **Geometry:** exact conductor, splitting, field degree, kernel degree, and class number.
- **Construction:** an observed endpoint, census, kernel, map, or dual under a stated resource envelope.
- **Native action:** the best certified action available directly on each endpoint.
- **Transfer advantage:** the base-two log ratio of incumbent to fully charged candidate cost under a declared target count and common cost unit.

For a declared target count `H`, the bit-valued transfer estimand is

$$A(H) = \log_2\left(\frac{C_{\mathrm{inc}}(H)}{C_{\mathrm{discover}} + C_{\mathrm{construct}} + C_{\mathrm{certify}} + H(C_{\mathrm{map}} + C_{\mathrm{solve}} + C_{\mathrm{recover}} + C_{\mathrm{verify}}) + C_{\mathrm{failures}}}\right).$$

Positive `A(H)` means the fully charged candidate is cheaper. Costs must use one declared primitive or calibrated unit; a degree, byte count, or field dimension cannot be inserted as if it were runtime.

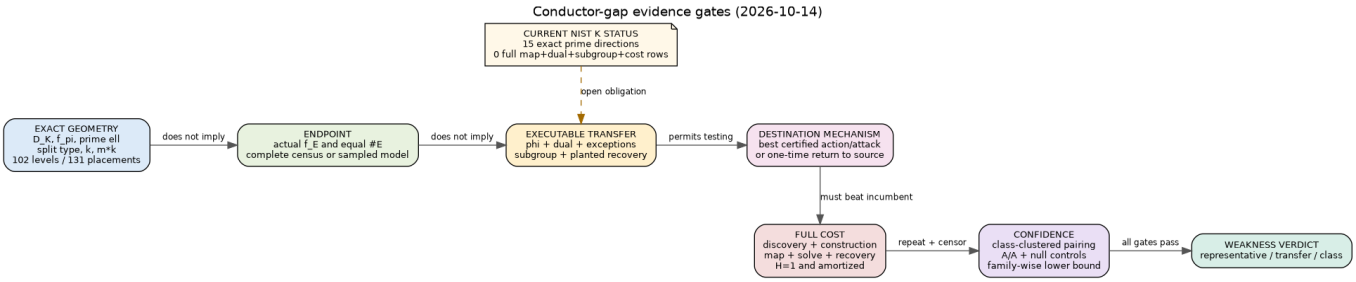



Figure 2: Exact conductor geometry is only the first of six gates required for a weakness conclusion.

Editable vector: `evidence-flow.svg`; source: `evidence-flow.dot`.

## NIST K curves

The table reports the cheapest direct kernel direction by absolute extension degree for each named K curve. Values are exact structural derivations. The status column is the strongest archived endpoint evidence; none includes the complete point-map/dual/transfer package.

| curve | cheapest `ell` | type | `k` | absolute `m*k` | kernel degree | packed bytes | endpoint status |
|---|---|---|---|---|---|---|---|
| K-163 | 45,641 | split | 20 | 3,260 | 22,820 | 464,978 | 6 endpoints; map not retained |
| K-283 | 1,697 | inert | 848 | 239,984 | 848 | 30,034 | census implementation only; not run canonically |
| K-571 | 3,952,463 | split | 3,461 | 1,976,231 | 1,976,231 | 141,053,559 | structural only |
| K-233 | 5,610,641 | split | 12,040 | 2,805,320 | 2,805,320 | 81,704,975 | structural only |
| K-409 | 21,262,439,877,311 | inert | 2,126,243,987,731 | 869,633,790,981,979 | 10,631,219,938,655 | 543,521,119,363,788 | structural only |

The full exact factorizations are:

```
K-163: 45641 * 82153 * 8610311 * 56498081
K-233: 5610641 * 85310626991 * 150532234816721999
K-283: 1697 * 162254089 * 17791432075516525848369952866271
K-409: 21262439877311 * 22431439539154506863
       * 57030553306655053533734286593
K-571: 3952463
       * 902118413539623892438989194519267681451899364508980776927700984910373365482803 9
```

K-163 has the most developed retained construction path. The archive retains six degree-45,641 endpoints at 61–87 seconds each, six degree-82,153 endpoints at 842–989 seconds each, and six conditional degree-82,153 descents from the 45,641 floor with inner isogeny timers of 828–1,019 seconds plus a separately logged 29-second embedding per run. These timings construct endpoint coefficients; they are not complete wall-clock or point-map evaluation timings. This round repaired the new degree-45,641 GP driver's silent startup exit and verified that it parses through the expensive `ffembed` stage under a bounded diagnostic. The full forward-map producer has not completed, and no map artifact is claimed.

K-283 has the smallest degree but not the smallest kernel field. Its degree 1,697 kernel polynomial would contain only about 30 KB of densely packed base field coefficients, yet the direct point construction lives over absolute degree 239,984. The CM census can identify endpoints without supplying the map. Its archived implementation remains unexecuted through the canonical runner.

K-409's smallest direction is already far beyond the current direct route: the derived packed kernel payload is about 543.5 TB (decimal), before language or algebra-system overhead, and its point field has absolute degree about `8.70e14`. This is a structural burden calculation, not a bounded failure or an impossibility proof.

## Other standardized curves and controls

The NIST scan gives exact `f_pi=1` for P-192, P-256, P-384, P-521, B-163, B-233, B-283, and B-409. `sect113r1` also has fundamental Frobenius discriminant and `f_pi=1`. These are zero-vertical-gap controls: they rule out this particular vertical-conductor mechanism, not horizontal isogenies, special endpoint models, implementation defects, or other attacks.

P-224 has exact `f_pi=3`, with the prime ramified in its CM field. Four rational linear degree-3 kernels and codomains have been enumerated, but the named curve's actual `f_E` remains unresolved in `{1,3}` and the edge directions, closure, point map, and dual are not certified. It is therefore a potential direction, not one of the 43 certified Koblitz gap rows.

B-571 remains unresolved in the archived NIST scan and is not silently counted as an `f_pi=1` control. The retained partial-factorization bound is still useful: `|D_K| >= 4.114e46` (164.82 bits), implying no non-scalar endomorphism below degree `1.029e46` under that archived derivation.

The published full `sect113r1` discrete-log computation is a real weak-size calibration, but it is not evidence for a conductor-gap mechanism: this panel computes no vertical conductor direction for that curve.

**Horizontal-scope status:** zero horizontal directions were enumerated or timed in this round. In particular, `f_pi=1` rules out a vertical conductor descent, not horizontal isogenies within the maximal-order class. Horizontal endpoint/action discovery remains a separate unmeasured requirement for the P, B, and `sect113r1` controls.

## What changes at each measured volcano level

The strongest same-action evidence is the accepted 160,000-solve `m=37/m=41` replication. Each number is `log2(child mean / parent mean)` in counted operations, so positive means more work on the child. These comparisons use native negation at both levels and do not charge an isogeny transfer.

| class | factor | other factor absent, bits [95% CI] | other factor present, bits [95% CI] |
|---|---|---|---|
| m=37 | 73 | +0.002917 [-0.020485,+0.026585] | -0.017605 [-0.038453,+0.003022] |
| m=37 | 2,663 | +0.008627 [-0.018443,+0.035801] | -0.011895 [-0.027377,+0.003650] |
| m=41 | 409 | -0.001430 [-0.049462,+0.044956] | -0.002297 [-0.040039,+0.036423] |
| m=41 | 1,721 | +0.016988 [-0.030100,+0.062547] | +0.016121 [-0.022858,+0.055663] |

All eight intervals contain zero. The three `m=37` level-versus-crater simultaneous 90% ratio intervals satisfy the frozen

+/-3% equivalence margin. The global six-level assertion is false because the `m=41` intervals are too wide, not because a nonzero effect was detected.

The accepted `m=51`, degree-271 comparison holds both levels to signed `tau^17`: `+0.005762506` bit with fixed-orbit 95% interval `[-0.032170483,+0.043374391]`. A separate later discovery scan certifies 42 bottom-to-parent endpoint pairs across `f_E=271*120871 -> f_E=120871`; it does not serialize a point map or dual.

The subsequent four-level `m=51` native-negation run completed all 20,480 solves. Its four local-edge point estimates range from -0.02635 to +0.02633 bit. However, the A and B held-out A/A intervals crossed the frozen +/-0.15 bit margin by 0.00304 and 0.00904 bit. The run is an intact observation with status `aa_alert_labels_withheld`; its diagnostic intervals cannot be used to claim equivalence, an effect, or absence of an effect.

These results directly address the proposed "large prime conductor gap" heuristic: in the measured same-action arms, adding the larger factors 2,663, 1,721, or 120,871 did not produce a resolved multi-bit shift. The evidence does not generalize to NIST-sized gaps or to a best-action endpoint comparison.

## Why the modeled 3–4 action bits matter, but do not establish weakness

For a prime-order Koblitz subgroup with a certified signed action of size `2m`, the ideal random-map collision discount relative to negation is

$$\frac{1}{2}\log_2(m) \text{ bits.}$$

The derived ideal values are 3.674 (K-163), 3.932 (K-233), 4.072 (K-283), 4.338 (K-409), and 4.579 (K-571); they are not measured NIST runtimes. The `m=37` and `m=41` proxy measurements observe 2.487 counted-operation bits (pointwise 95% interval `[2.454,2.521]`) and 2.685 bits (`[2.631,2.741]`) for native signed `tau` versus native negation, respectively. The floor-level same-negation differences are near zero by comparison.

Thus, the multi-bit signal is presently **action availability**, not conductor size. Descending from a Koblitz crater often removes a cheap self-action; that would make a floor less attractive for rho, unless the floor supplies a different certified attack or a transfer strategy whose complete cost beats the source's signed-Frobenius incumbent. A transported action

$$[d^{-1}] \circ \phi \circ \tau \circ \hat{\phi}$$

must charge the dual, source action, forward map, inverse-degree scalar, canonicalization, and conversions every time it is used. One-time return to the Koblitz source is a mandatory dominance control.

## Statistical confidence and weak-curve gate

Exact identities receive independent arithmetic verification, not p-values. For runtime, construction success, or solver work:

1. Use the hierarchy `isogeny class -> conductor edge -> parent orbit -> endpoint orbit -> target/process block`. Twists and Galois conjugates are matched or nested observations, not extra classes.
2. Pair source and candidate by planted scalar, seed, host, and process block. Freeze equal-curve or equal-orbit weighting before execution; also examine endpoint tails because a level mean can hide a rare weak representative.
3. For this purposively selected fixed panel, cluster or resample only the nested edge/orbit/target units and scope inference to these curves; do not treat the 20 class labels as population draws. A future cross-family claim must preregister its sampling frame and exchangeability assumptions. Use max-|T| simultaneous intervals (Holm as fallback), untouched endpoint orbits, and target streams to confirm any selected outlier.
4. Keep timeouts right-censored for time-to-certificate and failures for a fixed-budget success probability. Treat OOM as failure under that memory envelope; invalid certificates are failures, never censoring.
5. Preserve A/A, coordinate-isomorphism, same-action, positive-map, and zero-gap controls. A failed mandatory control withholds the dependent inference rather than erasing the observations.
6. Report cold-start `H=1`, the declared multi-target workload, and break-even `H`. Do not amortize discovery or construction without naming `H`. Size a future confirmatory run from a preregistered minimum detectable effect and target power before looking at candidate outcomes.

A weak-representative verdict requires exact endpoint identity, an applicable verified attack, full accounting, planted-scalar recovery, and a family-wise one-sided 95% lower bound above a material margin predeclared by that future experiment. A source-transfer verdict additionally requires public (or explicitly advice-scoped) map/dual reconstruction and subgroup recovery. A class-wide verdict requires a uniform invariant or construction over the declared class.

For a specifically predeclared three-bit reduction claim, the simultaneous lower bound on fully charged advantage must exceed **3 bits**. A 3–4 bit point estimate, ideal orbit formula, kernel degree, or conductor ratio is insufficient.

## Next discriminating executions

1. **K-163 / 45,641:** finish the repaired deterministic forward-map build; retain the full kernel polynomial and exceptional-input behavior. Add the algebraic dual, both composition identities, standard-subgroup transport, and planted recovery before benchmarking.
2. **m=37 / 73 positive control:** persist the existing native forward map and checked pullback as a complete map/dual certificate, then compare one-time return with the transported-action arm. This should fail closed before a full-size K-163 cost run is admitted.
3. **K-283 / 1,697:** first admit and canonically execute the existing CM census to certify the 1,698-model floor. Then stage the forward-map route as separate bounded gates: field embedding plus an exact-order torsion witness; enumeration and descent of the 848 half-kernel abscissae into a retained base-field polynomial; and codomain, exception, subgroup, planted-log, and homomorphism checks. Construct and certify the dual separately. Census success alone is not a transfer.
4. **Replication panel:** add independent conductor lattices, not merely more targets within `m=37`, `m=41`, or `m=51`. Use the exact 43-row panel for stratification by `log2 ell`, `log2(mk)`, and split type, with held-out classes for confirmation.

## Evidence and provenance

All quantitative statements above are `internal` repository evidence or derived from it. The builder records SHA-256 hashes for every controlling source in `panel.json`. That file is a structural panel plus selected archived effects, not an exhaustive performance warehouse; the linked source analyses remain authoritative for statistics not copied here.

- NIST identities and Frobenius conductors: internal scan.
- Binary conductor survey, CM floors, K-163 endpoints, and second level: internal survey archive.
- Four-level exact proxy replication: internal results.
- Accepted degree-271 result: internal analysis.
- Deep `m=51` discovery and gated native-rho observation.
- Open transfer obligations: Koblitz maps and P-224 maps.
- Independent factor, minimal-order, lattice, source-transcription, and gate checks: `verify_panel.py` and `verification.json`.

Primary background references:


- D. Kohel, *Endomorphism rings of elliptic curves over finite fields*, PhD thesis, UC Berkeley, 1996, https://hdl.handle.net/1721.1/10973.
- NIST SP 800-186, *Recommendations for Discrete Logarithm-based Cryptography: Elliptic Curve Domain Parameters*, 2023, https://doi.org/10.6028/NIST.SP.800-186.
- Standards for Efficient Cryptography Group, *SEC 2: Recommended Elliptic Curve Domain Parameters*, version 2.0, 2010, section 3.2.1 (`sect113r1`), https://www.secg.org/sec2-v2.pdf.
- G. Wiener and R. Zuccherato, *Faster Attacks on Elliptic Curve Cryptosystems*, SAC 1998, https://doi.org/10.1007/3-540-48892-8_15.
- E. Wenger and P. Wolfger, *Harder, better, faster, stronger: elliptic curve discrete logarithm computations on FPGAs*, J. Cryptographic Engineering 6 (2016), https://doi.org/10.1007/s13389-015-0108-z.


## Reproduction

```
MPLCONFIGDIR=/tmp/conductor-gap-panel-mpl \
XDG_CACHE_HOME=/tmp/conductor-gap-panel-cache \
```

```
python3 -B research/conductor_gap_panel_20261014/build_panel.py

python3 -B research/conductor_gap_panel_20261014/verify_panel.py

dot -Tsvg research/conductor_gap_panel_20261014/evidence-flow.dot \
  -o research/conductor_gap_panel_20261014/evidence-flow.svg

pandoc research/conductor_gap_panel_20261014/REPORT.md \
  --from=markdown+raw_tex+tex_math_single_backslash \
  --pdf-engine=xelatex \
  --resource-path=research/conductor_gap_panel_20261014 \
  -o research/conductor_gap_panel_20261014/report.pdf
```

This rebuilds derived views only. It does not rerun the archived scientific trials or promote any claim.